Heathcare Data Analysis Report

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Introduction:

This dataset presents a comprehensive collection of hospital patient record metadata, capturing core details such as patient name, gender, blood type, insurance information, medication, and more. In addition, it also provides identifiers such as the doctor's name, hospital, billing amount, test results and other attributes that are relevant for a health-care environment. Utilizing the metadata, this will enable analysis beyond the surface level revealing correlations between the identifiers and hospital operations when it comes to patients.

The central focus of this project is to identify and interpret patterns by using Python, SQL, and the visualization tool Power BI to answer questions such as:

- How do demographic factors such as gender and blood type relate to common treatments, lengths of stay, or medication regimens?
- Does a certain hospital have a different time length between admission and discharge date?
- Are there trends between blood types with certain medical conditions?

Through addressing these questions and future questions, this analysis seeks to deliver actionable insights into the functioning and optimization of patient care, administrative workflow, and whatever findings that are hoped to be unveiled through this analysis.
Ultimately, this report aims to elevate the understanding of hospital operations by transforming raw metadata into clear, actionable intelligence about patient needs, care quality, and institutional performance, fueling continuous improvement and innovative practice across the healthcare ecosystem.

For full report reference, click the link to direct you to the full findings and analysis for each programming language:

- Python

Python:

In the data cleaning portion, only the columns and duplicate entries had to be changed as it would give the data more clarity and less skewness when conducting analysis on Python and SQL. Additionally, data types had to be changed in order to make the columns viable for analysis, name changes are mentioned on the referenced Python analysis.

Following transforming and cleaning the data, correlations were conducted among the columns. Seeing what column affects another is crucial to understanding what can help in this specific healthcare environment, if not specific a general basis.

While doing a correlation visualization, I came with up with the idea to do 3 correlations, numeric, categorical and numeric + categorical respectively. The insights that were drawn out before further analysis were the following:

- No strong correlations, meaning there were no columns that had a stronger impact on the other.
- It was either no correlation or a small impact.
- There was a slight weak positive correlation with age and the amount of days a patient had to stay.
- The longer a patient stays, it would affect their bill.

After the correlation visualization, another series of visualizations was applied. This was in order to see if there were any anomalies, outliers in the dataset. After conducting these graphs, it was concluded that the data contained a normal distribution with no outliers, and everything was balanced meaning nothing was higher or lower than the other; everything was evenly split.

Following this, statistical testing was conducted on the eye-catching correlations, and then there were other statistical testing conducted to make sure there was nothing hidden from the visualizations.

There were 4 distinct statistical testings conducted with different types of tests between columns to determine validity of the correlations and appropriate testing. The following tests were conducted:

- Pearson correlation coefficient (Pearson's r)
- Welch's T-Test (Unequal variance t-test)
- One-way ANOVA
- Chi-Squared Test of Independence

Using these statistical testing methods, the following insights were uncovered:

- Against treatment days and age, there is no correlation between the two categories.
- Treatment days vs Billing amount, there is no correlation between the two.
- There is no difference in billing between the two genders.
- The average billing amount is the same across the admission types.
- There is no correlation between gender and medical condition.
- Between test results and insurance provider, there is no correlation between each other.
- There is no relationship between blood type and test results.

From these tests, it is shown that majority of the relationships are independent of each other and/or have a weak correlation with each other. Following this, one column does not affect another so everything can be focused as one entity instead of worrying about another category being affected.

The last segment of this analysis was clustering columns to see what useful insights would be gathered. In this process, 4 groups of clusters were conducted as there were relevant columns needed separately. The following cluster groups are as follows:

Patient Groups
Age
Treatment Days
Billing Amount

Clinical Severity
Age
Treatment Days
Various Medical Conditions
Test Results
Medications

Utilization & Cost
Treatment Days
Billing Amount
Room Number
Admission Types

Insurance & Payment
Billing Amount
Treatment Days
Various Insurance Providers

With these groups in mind, the following interpretations were taken for each of the clustered groups:

- Patient Groups:
  - Younger patients with longer treatment durations and lower-than-average billing amounts.
  - Very young patients with slightly shorter treatment durations and higher-than-average billing amounts.
  - Older patients with longer stays and higher-than-average billing amounts.
  - Older patients with short treatment durations and lower-than-average billing amounts.
- Clinical Severity:
  - The analysis revealed four distinct patient clusters, which were created with similarities in age and length of hospital stay.
  - Younger patients tended to have shorter hospital stays, whereas older patients stayed longer, but other factors like medical conditions or medications did not differ much between the groups.
  - Age and treatment duration were the main factor of patient grouping, while clinical and medication patterns remained largely similar across all groups.
- Utilization & Cost:
  - Patients were divided into four distinct groups based on treatment duration, billing data, and admission details to uncover shared patterns.
  - Two clusters consisted of individuals with extended hospital stays, while the other two reflected shorter treatment periods. Despite these differences in length of stay, billing amounts, room assignments, and admission types remained largely consistent across all groups.

  - Treatment duration emerged as the key factor distinguishing patient categories, with the other variables playing only a minor role in this cluster distribution.

- Insurance & Payment:
  - Patients were grouped into four categories based on treatment duration, billing amounts, and insurance coverage.
  - The clusters portrayed distinct patterns: one group had extended stays with lower costs, another had short, low-cost treatments, a third showed brief but high-cost care, and the last combined long stays with high expenses.
  - Overall, length of stay and billing amount were the main factors distinguishing the groups, while insurance provider distribution such as Blue Cross, UnitedHealthcare, and Medicare showed only minor variation across clusters.

From the results of my Python analysis, it was revealed the true relationships between columns, if there were any abnormalities within the data in which there wasn't as this dataset's entries was accurate enough to draw valuable results.

SQL: